COMSATS University Islamabad, Lahore Campus

Mid Term Examination – FALL 2025

Course Title:	Web Technologies			Course Code:	CSC336	Credit	3(2,1)
Course Instructor/s:	Asif Shahzad			Programme Name:	BSCS		
Semester:	5	Batch:	FA23	Section:	G1, G2	Date:	Nov. 6, 2025
Time Allowed:	**1.5 Hour**			**Maximum Marks:**	**50**		
Student's Name:				Reg. No.			

Important Instructions

1. Please implement functionality that is asked, do not add extra features. Write clean and minimum code.
2. It's suggested to first make logic on rough page at last page of your answer sheet, once its final, then write the answer to avoid cutting/overwriting.

OBJECTIVE PART

Time Allowed: 20 minutes

Question 1 *[CLO-1] Understanding (Classify web architectures and related applications.)* **15 Marks**

1 Which layer does HTTP belong to?
A. Network
B. Transport
C. Application
D. Session

2 Content-Type "application/x-www-form-urlencoded" is used when:
A. Uploading files
B. Sending simple form data
C. Sending binary data
D. Streaming video

3 When uploading a file through `multipart/form-data`, each part in payload is separated by:
A. Query string delimiters
B. Boundary string defined in header
C. Special encryption tag
D. A fixed byte pattern

4 If a POST request contain a binary file along with normal input fields. Where the content type information of the binary file would be stored in HTTP packet:

A. Only in file meta data i.e. binary stream

B. In HTTP packet headers e.g. content-type

C. Inside the packet body

D. Content type of file is not sent by browser

5 A major disadvantage of Microservices over Monolithic design is:

A. Difficult code reuse

B. Complexity in deployment and inter-service communication

C. Reduced scalability

D. Low fault isolation

6 Macroservices architecture is chosen over microservices mainly when:

A. Project scale is small but needs modularity

B. Each function must be independently deployable

C. Every feature has its own database

D. There are hundreds of tiny services causing overhead

7 URL encoding is used to:

A. Encrypt passwords

B. Represent unsafe characters

C. Shorten URLs

D. Compress data

8 DAO layer is responsible for:

A. UI design

B. Data persistence

C. Business logic

D. Security

9 Increasing tiers improves scalability but adds:

A. Performance loss only

B. Cost and complexity

C. Simplicity

D. Redundancy

10 Single Page Applications (SPAs) load SPB (Structure, Presentation, Behavior):

A. On every page reload

B. Once, and then fetch data via APIs

C. From database directly

D. Only via FTP

11 A main issue with DSPB combined approach is:

A. Slow execution

B. Difficult mobile app integration

C. Browser incompatibility

D. Low network speed

12 Decoupling data from SPB leads to:

A. Tight coupling

B. Maintainability and reuse

C. Increased cost

D. Manual updates

13 The difference between `PUT` and `POST` is that:

A. PUT creates multiple resources; POST only one

B. PUT is idempotent; POST is not

C. PUT cannot include a body

D. POST updates existing resources

14 Which HTTP header indicates the MIME type of the *response body*?

A. Content-Type

B. Accept

C. Server

D. Content-Length

15 A monolithic application converting into SOA is expected to improve:

A. UI design consistency only

B. Modularity and independent deployability

C. File storage performance

D. CPU scheduling

SUBJECTIVE PART

Time Allowed: 70 minutes

Question 2 *[CLO-2] Applying (Demonstrate the capabilities for developing professional front-end using client-side technologies.)* **15 Marks**

Create a responsive webpage layout with three columns using the provided HTML structure. Apply CSS (e.g. flexbox and media queries) to make the layout adapt to different screen sizes using the below given breakpoints. Do not apply basic styling such as colors, padding, margin, and borders.

1. Desktop view (width ≥ 992px): Display three columns side by side.
2. Tablet view (width between 600px and 991px): Display two columns per row.
3. Mobile view (width ≤ 600px): Display one column per row (stacked vertically).

```
<main class="container">
<section class="column"></section>
<section class="column"></section>
<section class="column"></section>
</main>
```

Question 3 *[CLO-2] Applying (Demonstrate the capabilities for developing professional front-end using client-side technologies.)* **20 Marks**

Create a **simple web page** that uses **HTML, CSS, and JavaScript/jQuery** to display weather information.

1. The page should include:
   - A **heading** such as "Weather Info"
   - A Form to get **longitude** and **latitude** values in input fields with labels.
   - A **button** labeled **"Load Weather"**
   - A **weather box or div** where the **temperature** and **wind speed** will be displayed
2. When user clicks the **"Load Weather"** button, use **AJAX** to call the following API: https://api.open-meteo.com/v1/forecast?latitude=33.7&longitude=73.1&current_weather=true
3. Extract and display the **temperature** and **wind speed** values from the returned JSON inside your weather box.
4. Handle errors gracefully and display a suitable message if the API call fails.
5. Suppose JQuery is already added and your script would be loaded in head tag of the page. Using jQuery is recommend but not required, you can use XMLHttpRequest object too.

Expected JSON Structure:

```
{
  "latitude":33.7, "longitude": 73.1,
  "current_weather": {
    "temperature": 24.5,
    "windspeed": 6.7,
    "winddirection": 150 }
}
```